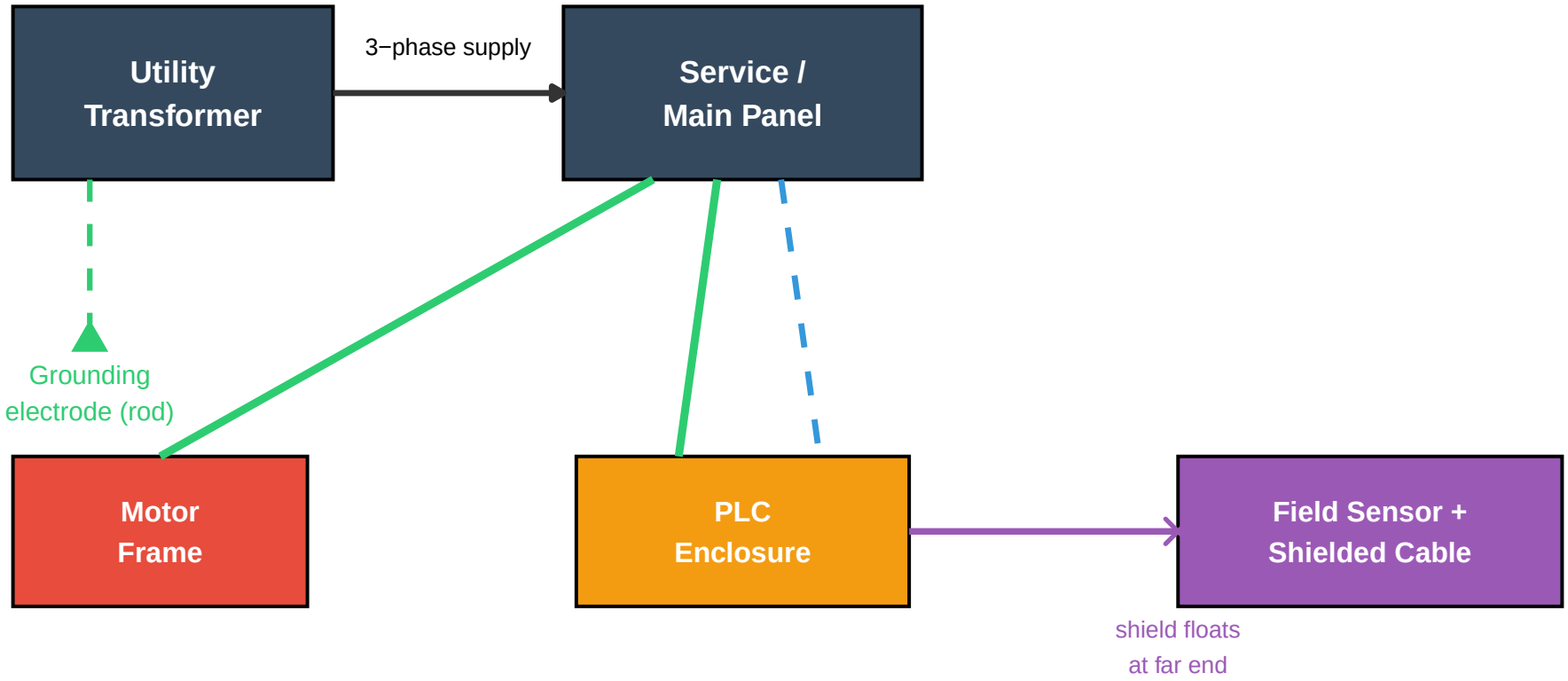
Four Conductive Paths, One Confusing Name

Equipment bonding, system grounding, functional/signal ground, and shield each serve a different purpose



- Equipment bonding −− low−impedance fault−current path (CEC Sec. 10 / NEC Art. 250)
- Functional / signal ground −− reference only, NOT a fault−current path
- Cable shield −− grounded at ONE end only (low−frequency instrumentation)
- System grounding electrode −− voltage reference / lightning−surge path to earth